

1. Introduction

This project was created for learning and exploratory purposes. The dataset was downloaded from the official NASA website and contains information from the Kepler space telescope mission, whose primary goal was to search for exoplanets – planets outside our Solar System. The discoveries were made using the transit method, which relies on the continuous observation of stars. If a cyclical dimming of a star's light was detected, it suggested the presence of a planet orbiting around it.

2. Physics Concepts

2.1 The Dataset

The dataset consists of 50 columns with features describing potential exoplanets. The project focuses mainly on ten of them:

- **a) koi_disposition** – classification of objects into categories based on their likelihood of being planets:
 - *candidate* – objects that are potentially planets,
 - *confirmed* – objects verified as planets,
 - *false positive* – objects mistaken for planets.
- **b) koi_period** – the orbital period of the object around its star (measured in Earth days).
- **c) koi_depth** – transit depth. It shows how much the star's brightness dropped when the object blocked it, measured in Parts Per Million (ppm).
- **d) koi_prad** – planet size (measured in Earth radii).
- **e) koi_impact** – describes how close the planet passes to the center of the star's disk during the transit.
- **f) koi_insol** – Insolation Flux – describes how much energy from the star reaches the planet (baseline value = 1, which represents the value of this parameter on Earth).
- **g) koi_snr** – Signal-to-Noise Ratio – the signal level emitted by the object.
- **h) koi_steff** – star temperature [K].
- **i) koi_slogg** – surface gravity of the star ($\log(g)$).
- **j) koi_srad** – star radius.

2.2 Key Physics Concepts Used in the Project

- **a) Fulton Gap** – used to establish the range for rocky planets. According to NASA, planets are divided into those with a radius equal to or smaller than Earth's radius [R], and those up to 2 times larger. The larger ones are considered planets that rapidly attract an atmosphere composed of hydrogen and helium, growing into gas planets. The Fulton Gap describes a deficit in the number of existing planets with a radius between $1.5R$ and $2R$. On the Power BI page describing potentially habitable planets, this gap was used to classify planets as rocky.
- **b) Synchronous Rotation (Tidal Locking)** – this is a limit calculated by astrophysicists describing a phenomenon where a planet's rotation around its own axis becomes synchronized with its orbit around the star. This results in a day on the planet being equal to its orbit around

the star (1 day = 1 year). Consequently, one side of the planet would face the sun all year round, while the other side would never experience sunlight.

- **c) Insolation Flux** – the amount of energy a planet receives from its star. It is an indicator where a value of 1.0 means the planet gets exactly the same amount of light as Earth.
- **d) Habitable Zone** – based on the collected data, information, and laws of physics, key parameters were determined to potentially identify a new planet suitable for life. These are:
 - Star surface gravity within the range of 4 to 4.7,
 - Orbital period of the object around the planet,
 - Insolation Flux within the range of 0.3 to 4,
 - Planet size within the range of 0.5 to 2,
 - Planet surface temperature from 180 K to 310 K.

3. Results

3.1 Python

The Python language (along with the Pandas, NumPy, Matplotlib, and Seaborn libraries) was used at the very beginning of the project to conduct the initial EDA and clean the dataset structure. Columns where missing values reached 100% – `koi_teq_err1` and `koi_teq_err2` – were removed. In the `kepler_name` column, a significant amount of missing data was detected (75%), but it was consciously not removed. It plays a key informational role, allowing for the identification of named confirmed planets on the final dashboard. Python was also used to find correlations between astrophysical parameters. This allowed for the quick identification of the strongest relationships and optimized subsequent work in Power BI.

3.2 SQL

An interesting result in SQL was highlighted by this code snippet:

```
select kepler_name as `Name`, koi_model_snr as `SNR`, koi_prad as `Size of planet`,  
koi_steff as `Star temperature K`, koi_insol as `Insolation Flux`, koi_teq as  
`Planet temperature`  
from `kepler-497213.Keppler_data.raw_kepler_data`  
where koi_disposition = 'CONFIRMED'  
order by koi_model_snr desc  
limit 10;
```

It shows objects classified as CONFIRMED (existing exoplanets) and their key parameters. The median SNR for planets with a CONFIRMED status was 30.6, and the average was 87.92. However, in these specific cases, the highest values reached almost 6,000, meaning that these types of planets distorted the average result. Planets with such a high emitted signal are characterized by extreme temperatures and sizes, making them unsuitable for life.

Wiersz	Name	SNR	Size of planet	Star temperature K	Insolation Flux	Planet temperature	Status
1	Kepler-2 b	5945.9	16.1	6440.0	4148.92	2048.0	CONFIRMED
2	Kepler-1 b	4304.3	13.04	5820.0	761.46	1339.0	CONFIRMED
3	Kepler-13 b	4061.9	21.42	9107.0	37958.27	3559.0	CONFIRMED
4	Kepler-12 b	3535.6	18.05	5953.0	757.65	1338.0	CONFIRMED
5	Kepler-6 b	3003.8	13.06	5640.0	797.62	1356.0	CONFIRMED
6	Kepler-423 b	2833.1	12.73	5561.0	603.57	1265.0	CONFIRMED
7	Kepler-7 b	2590.1	16.66	6036.0	1182.79	1496.0	CONFIRMED

Another interesting part of the analysis was creating a ranking of the top 20 objects with the strongest signal separately for each type of star, while simultaneously calculating the average SNR for each group. Among other things, this aimed to compare and point out outliers within individual categories.

Code:

```
with Imp_classification as (

select kepid, kepler_name, koi_disposition, koi_model_snr,
case
  when koi_steff between 5200 and 6000 then "Yellow dwarf"
  when koi_steff between 3500 and 5200 then "Orange dwarf"
  when koi_steff <= 3500 then "Red dwarf"
  else "Unkown"
end as `Star clasification`
from `kepler-497213.Kepler_data.raw_kepler_data`

),

ranking_raport as(

  select
    kepler_name,
    `Star clasification`,
    koi_model_snr as `SNR`,
    rank() over (partition by `Star clasification` order by koi_model_snr desc) as
    planetary_rank,
    round(avg(koi_model_snr) over(partition by `Star clasification`),2) as `Average
    SNR`
  from Imp_classification
  where koi_disposition = "CONFIRMED"

)

SELECT * from ranking_raport
where planetary_rank <= 20
```

```
order by `Star clasification` asc;
```

Part of result:

Wiersz	kepler_name	Star clasification	SNR	planetary_rank	Average SNR
1	Kepler-425 b	Orange dwarf	1549.0	2	62.62
2	Kepler-45 b	Orange dwarf	1341.6	3	62.62
3	Kepler-785 b	Orange dwarf	1252.5	4	62.62
4	Kepler-486 b	Orange dwarf	1126.2	5	62.62
5	Kepler-489 b	Orange dwarf	871.4	6	62.62
6	Kepler-3 b	Orange dwarf	862.3	7	62.62
7	Kepler-695 b	Orange dwarf	568.2	8	62.62
8	Kepler-167 e	Orange dwarf	544.2	9	62.62
9	Kepler-816 b	Orange dwarf	343.4	10	62.62
10	Kepler-94 b	Orange dwarf	312.7	11	62.62

3.3 Power BI

The final stage of the project was to find planets that could potentially be considered habitable. For this purpose, a funnel was created using filters (measures written in the DAX language and the parameters described in section 2.2 d)), which allowed for the discovery of potential new planets suitable for life.

In Table 1, there is a planet named Kepler-452b that deserves special attention. It is characterized by parameters similar to the conditions found on Earth:

- a) Its radius is 1.09 of Earth's radius.
- b) A year on the planet lasts 384 days.
- c) It orbits a yellow dwarf – a sun-like star.
- d) The star's gravity is 4.58 (the sun's gravity is 4.44).
- e) However, it is colder – the temperature is 220K, which corresponds to -53 degrees Celsius.

CANDIDATES FOR NEW EARTH							
kepler_name	Planet Radius (Earth = 1)	Year (Days)	Star Type	Star Gravity	Insolation Flux	Planet Temperature (K)	Status
Kepler-62 f	1,43	267,28	Orange dwarf	4,65	0,44	207,00	CONFIRMED
Kepler-452 b	1,09	384,85	Yellow dwarf	4,58	0,56	220,00	CONFIRMED
Kepler-283 c	1,87	92,75	Orange dwarf	4,68	0,78	240,00	CONFIRMED
Kepler-1544 b	1,69	168,81	Orange dwarf	4,62	0,80	241,00	CONFIRMED
Kepler-442 b	1,30	112,30	Orange dwarf	4,68	0,79	241,00	CONFIRMED
Kepler-62 e	1,72	122,39	Orange dwarf	4,65	1,24	269,00	CONFIRMED
Kepler-69 c	1,73	242,47	Yellow dwarf	4,40	1,59	286,00	CONFIRMED
Kepler-1450 b	1,94	54,51	Orange dwarf	4,67	2,13	308,00	CONFIRMED
Kepler-437 b	1,56	66,65	Orange dwarf	4,63	2,14	308,00	CONFIRMED

Table 1. Candidates for new earth

